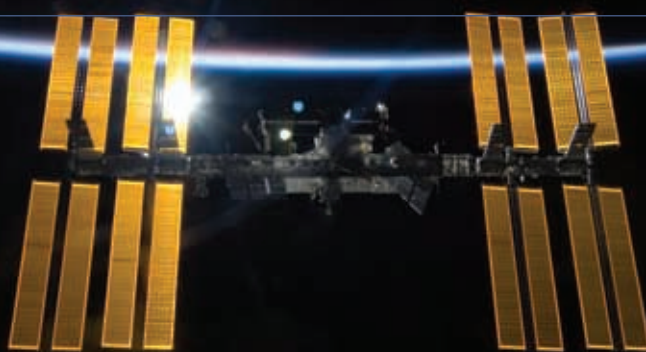




The impossible frontier

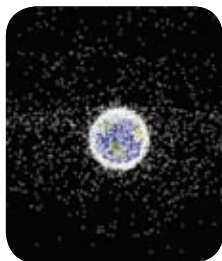
The final frontier may be the most exciting, and a trillion, trillion, trillion (*ad nauseam*) times larger than anything we've ever explored, but can we?

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Those of us from the Trekkie generation might have spent all our lives dreaming about going into space, and travelling to a galaxy far, far away. But space travel isn't even remotely close to as easy as popular sci-fi has made us believe.

SPACE JUNK

Although we've spoken a little about it before, we still have people asking how a tiny speck of dust or shrapnel that's the size of a pin can really harm us, let alone a spaceship. The problem is that we think of these things in very earth-bound terms. Those of us who have seen a car accident, even low speed ones, will know how easily metals buckle under stresses of a collision.



However, it's best if you think of a little drop of water – falling from a shower, you can't even feel it much, however, when falling from the sky, you feel a raindrop pricking you more. When you're riding a bike, the drops can sting like hell. Now speed up the raindrop to orbital speeds, and the rain drop will not sting, it will go through you, leaving a hole.

THE SHEER DISTANCE

To get an idea of the enormity of the great black void consider this: On a lucky day like 25th of August 2003, when one of our nearest planetary neighbour Mars was the closest that it had ever been in the last 60,000 years, the distance was still 56 million kilometers. But that's just a nearby rock. The real good stuff lies much farther; in other solar systems and other galaxies. Going to those celestial bodies would be actual 'exploration'. Our



nearest star, Proxima Centauri is about 4.3 light years away. In regular units that's a smudge under 43 trillion kilometers! Let's write it in numbers for you to marvel at the zeros - 43,000,000,000,000. If we were to miraculously solve the other issues of space travel and decided to undertake this journey using the Saturn V rockets we used to get to the moon, how long would it take? We did the math and found out for you – close to a staggering 1,94,927 years!

FRAGILE LITTLE HUMANS

While we may have mastered our earthly domain (just about) space is a very hostile environment. To sustain human life in space, a very controlled environment is needed. We need oxygen, water, and a near endless supply of food. To set up self contained ecosystem in a space vessel will be near impossible. Think of the soil, arti-



105 DAYS IN SPACE

An experiment called Mars 500, will test how human body will react to prolonged stays in a spacecraft, confined astronauts for 100 days. Here's a first hand account from the participants, who will be done with this only in November 2011!
<http://goo.gl/1eQne>

PREPARE TO FEEL SMALL

HD Timelapse footage of the VLT (Very Large Telescope) at the Atacama desert, Chile shows you a spectacular view of the know universe. The music and the spectacular visual display will not just move you, but will also make you wonder
<http://bit.ly/m6Y2YN>

SPACESHIP TWO

When the world's most flamboyant businessman decides to make a spacecraft, you take note. Richard Branson, founder Virgin group, discusses SpaceShip Two, Virgin Galactic's newest vessel and the world's first commercial spacecraft.
<http://goo.gl/irLhT>

ZERO GRAVITY IN SIN CITY

For \$5,000 you can experience zero gravity in Las Vegas. And no we aren't talking about substance abuse. Commercial weightless flights take passengers aboard a modified Boeing 727 to experience about a fleeting glimpse of weightlessness
<http://goo.gl/3xDpB>